ITI0402 Loogika, sügis 2026
Harjutus 10
Lahendused

1.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(f(x')) \to p(f(x'))}\ \text{hyp.}}{\forall x\ p(x) \to p(f(x'))}\ \forall\mathcal{L}}{\forall x\ p(x) \to \forall x\ p(f(x))}\ \forall\mathcal{R}}{\to \forall x\ p(x) \supset \forall x\ p(f(x))}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(y', f(y')) \to p(y', f(y'))}\ \text{hyp.}}{p(y', f(y')) \to \exists z\ p(y', z)}\ \exists\mathcal{R}}{\forall x\ p(x, f(x)) \to \exists z\ p(y', z)}\ \forall\mathcal{L}}{\forall x\ p(x, f(x)) \to \forall y\ \exists z\ p(y, z)}\ \forall\mathcal{R}}{\to \forall x\ p(x, f(x)) \supset \forall y\ \exists z\ p(y, z)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x') \to p(x')}\ \text{hyp.}}{p(x') \to \exists x\ p(x)}\ \exists\mathcal{R}}{\neg\exists x\ p(x), p(x') \to}\ \neg\mathcal{L}}{\neg\exists x\ p(x) \to \neg p(x')}\ \neg\mathcal{R}}{\neg\exists x\ p(x) \to \forall x\ \neg p(x)}\ \forall\mathcal{R}}{\to \neg\exists x\ p(x) \supset \forall x\ \neg p(x)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{p(a) \to p(a)}\ \text{hyp.}}{p(a) \to \exists x\ p(x)}\ \exists\mathcal{R} \quad \dfrac{\dfrac{}{p(b) \to p(b)}\ \text{hyp.}}{p(b) \to \exists x\ p(x)}\ \exists\mathcal{R}}{p(a) \vee p(b) \to \exists x\ p(x)}\ \vee\mathcal{L}}{\to p(a) \vee p(b) \supset \exists x\ p(x)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x') \to p(x'), q(x')}\ \text{hyp.} \quad \dfrac{}{q(x') \to p(x'), q(x')}\ \text{hyp.}}{p(x') \vee q(x') \to p(x'), q(x')}\ \vee\mathcal{L}}{p(x') \vee q(x') \to p(x'), \exists x\ q(x)}\ \exists\mathcal{R}}{p(x') \vee q(x') \to \exists x\ p(x), \exists x\ q(x)}\ \exists\mathcal{R}}{\exists x\ (p(x) \vee q(x)) \to \exists x\ p(x), \exists x\ q(x)}\ \exists\mathcal{L}}{\exists x\ (p(x) \vee q(x)) \to \exists x\ p(x) \vee \exists x\ q(x)}\ \vee\mathcal{R}}{\to \exists x\ (p(x) \vee q(x)) \supset \exists x\ p(x) \vee \exists x\ q(x)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x') \to p(x')}\ \text{hyp.}}{p(x') \to \exists x\ p(x)}\ \exists\mathcal{R}}{\to \exists x\ p(x), \neg p(x')}\ \neg\mathcal{R}}{\to \exists x\ p(x), \forall x\ \neg p(x)}\ \forall\mathcal{R}}{\to \exists x\ p(x) \vee \forall x\ \neg p(x)}\ \vee\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x') \to p(x')}\ \text{hyp.}}{\to \neg p(x'), p(x')}\ \neg\mathcal{R}}{\to \exists x\ \neg p(x), p(x')}\ \exists\mathcal{R}}{\to \exists x\ \neg p(x), \forall x\ p(x)}\ \forall\mathcal{R}}{\to \exists x\ \neg p(x) \vee \forall x\ p(x)}\ \vee\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x'), p(x'') \to p(x''), \forall x\ p(x)}\ \text{hyp.}}{p(x') \to p(x''), p(x'') \supset \forall x\ p(x)}\ \supset\mathcal{R}}{p(x') \to p(x''), \exists x\ (p(x) \supset \forall x\ p(x))}\ \exists\mathcal{R}}{p(x') \to \forall x\ p(x), \exists x\ (p(x) \supset \forall x\ p(x))}\ \forall\mathcal{R}}{\to p(x') \supset \forall x\ p(x), \exists x\ (p(x) \supset \forall x\ p(x))}\ \supset\mathcal{R}}{\to \exists x\ (p(x) \supset \forall x\ p(x))}\ \exists\mathcal{R}$$